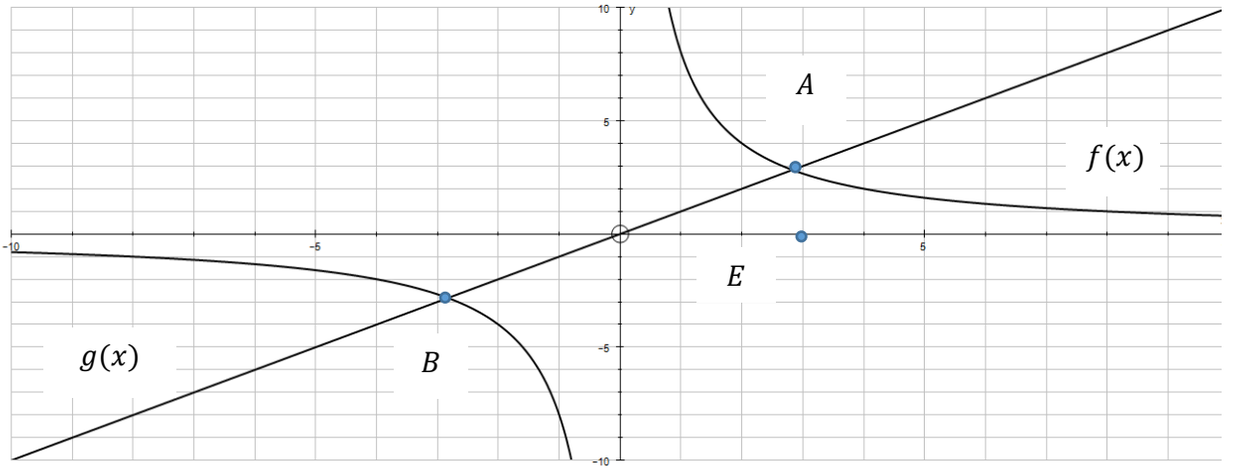
Revision 4

1. GOT IT! [15 MIN]

Example:

1. Given: $f(x) = \frac{8}{x}$ and $g(x)$ is a straight line.



1.1 Find the equation of the straight line $g(x)$

1.2 Determine the coordinates of the points A and B.

1.3 Determine the length of OE if $x \neq 3$

1.4 Give the domain of $f(x)$.

1.5 Give the equation of the axis of symmetry of $f(x)$.

Answer:

1.1 $g(x) = x$

It is important to remember that the graph is positive and that the $y - intercept$ is $(0; 0)$

$\therefore y = x$

1.2 $f(x) = g(x)$

$$\frac{8}{x} = x$$

$$8 = x^2$$

$$\pm\sqrt{8} = x$$

Point A and B are intersecting.

$\therefore$ The equations are equal to one another

$$f(\sqrt{8}) = \frac{8}{\sqrt{8}} = \frac{8}{\sqrt{8}} \times \frac{\sqrt{8}}{\sqrt{8}} = \frac{8\sqrt{8}}{8} = \sqrt{8}$$

Rationalise

$$f(-\sqrt{8}) = -\frac{8}{\sqrt{8}} = -\frac{8}{\sqrt{8}} \times \frac{\sqrt{8}}{\sqrt{8}} = -\frac{8\sqrt{8}}{8} = -\sqrt{8}$$

$\therefore A = (\sqrt{8};\ \sqrt{8})$ and $B = (-\sqrt{8};\ -\sqrt{8})$

1.3 $\sqrt{8} - 0 = \sqrt{8}$

$\therefore OE = \sqrt{8}$ units

To determine the distance, we subtract the x- values.

1.4 $x \in R; x \neq 0$

1.5 $y = x \ and \ y = -x$

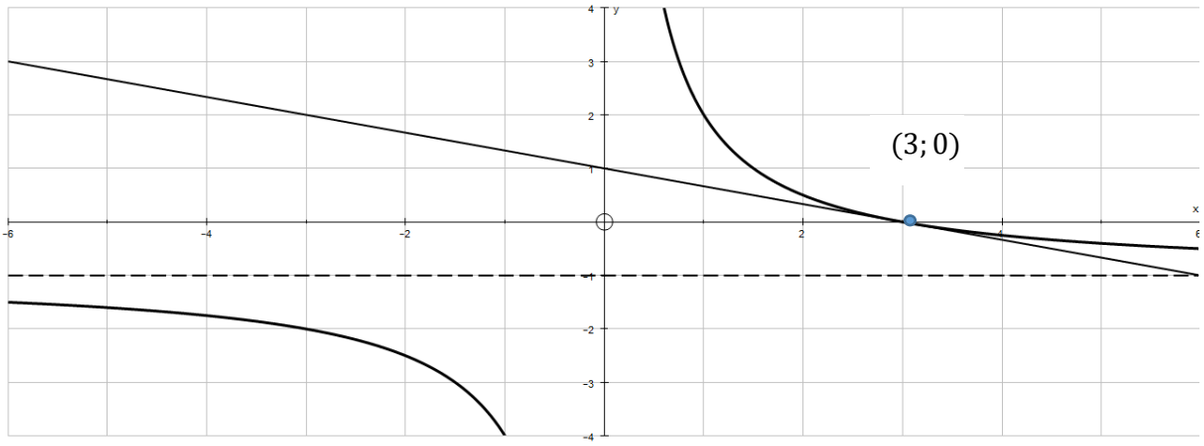
Symmetry line:

The line that you reflect over is called the line of symmetry. A line of symmetry divides a figure into two mirror-image halves.

2. GO AHEAD [30 MIN]

2.1 Study and use the diagram below to answer the following questions:

Given: $f(x) = \frac{a}{x} + q$ and $g(x) = mx + c$



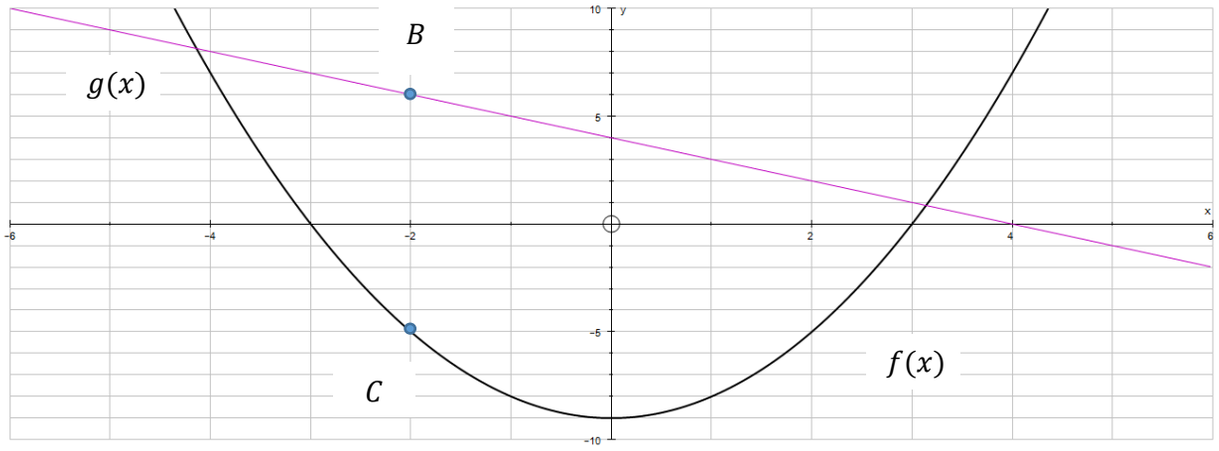
2.1.1 Write down the equation of $f(x)$ and $g(x)$.

2.1.2 Write down the range of $f(x)$.

2.1.3 Write down the domain of $g(x)$.

2.1.4 Determine the equation of $h(x)$ if $h(x) = -f(x) + 3$

2.2 The function of $f(x) = x^2 - 9$ and $g(x) = -x + 4$ is sketch below.



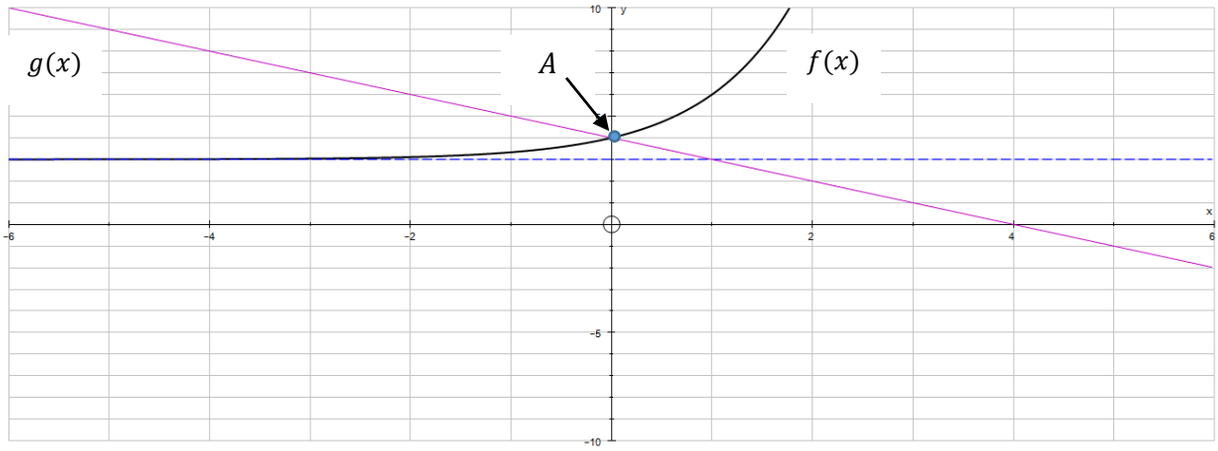
2.2.1 Write down the turning point of $f(x)$

2.2.2 Write down the range of $f(x)$.

2.2.3 Determine the length of BC.

2.2.4 For which value(s) of x is $f(x) < 0$

2.3 The function of $f(x) = 3^x + 3$ and $g(x) = -x + 4$ is sketch below. Point A is the intersection of $f(x)$ and $g(x)$



2.3.1 Write down the asymptote of $f(x)$.

2.3.2 Write down the range of $f(x)$.

2.3.3 Determine the coordinates of A.

2.3.4 For which value(s) of x is $f(x) > g(x)$